

A Tree-Metric Obstruction to Finite Weak Linear Rigidity

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Status. This is a partial result for Question 5 of Melleray–Petrov–Vershik, *Linearly rigid metric spaces and the embedding problem*, arXiv:math/0611049. It is not a full answer to the arbitrary finite-metric question.

Source question. In Section 3.3, after defining weakly linearly rigid spaces as those for which the Hausdorff–Kuratowski and Kantorovich–Rubinstein norms coincide, the authors ask:

Does there exist the finite WLR metric space with more than four points?

The criterion of linear rigidity says that any extremal Lipschitz function on X is almost realized as a function $\rho(x, \cdot)$. The only difference is the absence of $\pm$. Recall that if we take a supremum not by the set of functions $\pm\rho(x, \cdot)$, but by the set of functions $\frac{1}{2}(\rho(x, \cdot) - \rho(y, \cdot))$, then the coincidence of the corresponding norm with the maximal (KR) norm implies linear rigidity. But this difference is quite essential. There are some spaces which are weakly linearly rigid but not linearly rigid. We do not have any less or more complete description of such spaces. So only some examples follow.

1. Any metric space on three points gives a WLR-space (but it is never linearly rigid, as any finite metric space on more than two points, see theorem 3).

2. Let us define a family of WLR-spaces on four points. Let points be A, B, C, D ; let a, b, c be arbitrary positive numbers and define: $\rho(D, A) = a$, $\rho(D, B) = b$, $\rho(D, C) = c$, $\rho(A, B) = a+b$, $\rho(B, C) = b+c$, $\rho(A, C) = a+c$. It is easy to see that any extremal Lipschitz function is realized as $\pm\rho(X, \cdot)$ for some $X \in \{A, B, C, D\}$.

Question 5. *Does there exist the finite WLR metric space with more than four points?*

It looks more likely that there are no such spaces at least with sufficiently many points.

3. In the same time there are an infinite WLR-space which is not linearly rigid either. Consider the Urysohn space $\mathbb{U}$ and fix a point $a \in \mathbb{U}$. Add a point a' to the space $\mathbb{U}$ and define the distances $\rho(a', x) = \rho(a, x) + 1$ for any $x \in X$ (in particular, $\rho(a, a') = 1$). This new space $U' = \mathbb{U} \cup \{a'\}$ is WLR, but not linearly rigid.

Definitions. For a finite metric space X , write $\text{Lip}_0(X)$ for the space of real-valued Lipschitz functions modulo constants, represented by fixing one base point to have value 0. Its unit ball is

$$B_{\text{Lip}_0(X)} = \{f : |f(x) - f(y)| \leq \rho(x, y) \text{ for all } x, y \in X\} / \mathbb{R}.$$

In the finite case, X is WLR precisely when every extreme point of this unit ball is represented by a signed distance function

$$f(\cdot) = \pm \rho(x, \cdot) + c$$

for some $x \in X$ and some constant c . This is the finite exact form of the criterion stated in the source paper.

Proof Intuition. For a finite tree metric, the Lipschitz inequalities along the tree edges are independent coordinates. Once a root is fixed, a Lipschitz function is determined by its increments along the $n - 1$ edges, each lying in an interval $[-\ell(e), \ell(e)]$. Thus the Lipschitz unit ball is a box with 2^{n-1} vertices. Signed distance functions supply at most $2n$ of those vertices. When $n \geq 5$, there are too many box vertices, so at least one extremal Lipschitz function cannot come from a signed distance function.

Lemma 1. *Let $T = (V, E)$ be a finite tree with positive edge lengths $\ell(e)$, and let ρ be the induced path metric on V . Then $B_{\text{Lip}_0(V)}$ is affinely isomorphic to the box*

$$\prod_{e \in E} [-\ell(e), \ell(e)].$$

In particular, it has exactly $2^{|V|-1}$ extreme points.

Proof. Fix a root $o \in V$, orient every edge away from o , and represent $\text{Lip}_0(V)$ by functions satisfying $f(o) = 0$. The map

$$f \mapsto (f(v) - f(u))_{e=(u,v) \in E}$$

is a linear isomorphism from this quotient model onto $\mathbb{R}^E$, because the value of f at any vertex is recovered by summing increments along the unique path from o .

The Lipschitz condition on adjacent vertices gives

$$|f(v) - f(u)| \leq \ell(uv) \quad (uv \in E).$$

Conversely, if these edge inequalities hold and $x, y \in V$, then along the unique path $x = x_0, x_1, \dots, x_k = y$,

$$|f(x) - f(y)| \leq \sum_{i=0}^{k-1} |f(x_{i+1}) - f(x_i)| \leq \sum_{i=0}^{k-1} \ell(x_i x_{i+1}) = \rho(x, y).$$

Thus the edge inequalities are exactly the full Lipschitz unit-ball constraints. The stated box description and vertex count follow. $\square$

Theorem 1. *Let T be a finite weighted tree and let ρ be its path metric on the vertex set $V(T)$. If $|V(T)| \geq 5$, then $(V(T), \rho)$ is not weakly linearly rigid.*

Proof. By Lemma 1, the finite Lipschitz unit ball has $2^{|V|-1}$ extreme points. If (V, ρ) were WLR, each of these extreme points would have to be represented by a signed distance function $\pm \rho(x, \cdot) + c$. There are at most $2|V|$ such functions modulo constants. For $|V| \geq 5$,

$$2^{|V|-1} > 2|V|.$$

Hence at least one extreme point of $B_{\text{Lip}_0(V)}$ is not a signed distance function, contradicting the WLR criterion. $\square$

Theorem 2. *Among four-point weighted tree metrics on the tree vertices, the WLR spaces are exactly the weighted tripods. Equivalently, the underlying tree must be the star $K_{1,3}$.*

Proof. There are only two tree shapes on four vertices: the path P_4 and the star $K_{1,3}$.

For the star $K_{1,3}$, let D be the center and A, B, C the leaves, with edge lengths $a, b, c > 0$. Then

$$\rho(D, A) = a, \quad \rho(D, B) = b, \quad \rho(D, C) = c,$$

and the leaf distances are

$$\rho(A, B) = a + b, \quad \rho(A, C) = a + c, \quad \rho(B, C) = b + c.$$

This is precisely the four-point WLR family recorded in the source paper.

For the path P_4 , choose an ordering of the vertices

$$x_1 - x_2 - x_3 - x_4$$

with edge lengths $a, b, c > 0$. By Lemma 1, every choice of signs on the three edge increments gives an extreme point. Consider the alternating increment pattern

$$f(x_2) - f(x_1) = a, \quad f(x_3) - f(x_2) = -b, \quad f(x_4) - f(x_3) = c.$$

If this extreme point were a signed distance function, then its signs along the path edges would have to be those of the distance from some vertex, or the negative of such a distance. Along a path, a distance function has edge signs which change direction at most once: the signs point away from the root vertex; its negative points toward the root vertex. The pattern $+, -, +$ changes direction twice, so it is neither a distance function nor the negative of one. Therefore P_4 is not WLR. $\square$

Limitations. Theorem 1 rules out a large and natural class of finite metrics, including all finite weighted graph-path trees on at least five vertices. It does not rule out arbitrary finite metrics with five or more points. The full Question 5 remains open after this packet.

Verification. The script `code/check_tree_wlr.py` verifies the sign-pattern count for all labeled trees up to six vertices and checks the four-point path/star dichotomy.

References.

- J. Melleray, F. V. Petrov, and A. M. Vershik, *Linearly rigid metric spaces and the embedding problem*, arXiv:math/0611049.